Muhammad Ahzam

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EDUCATION

Institute of Business Administration (IBA), Karachi Expected 2027
BS in Computer Science
Coursework: Data Structures, Design and Analysis of Algorithms, OOP, Database Systems, Software Engineering, Web-Based App Development, Cloud Computing, Operating Systems, Computer Architecture and Assembly Language, Discrete Math, Machine Learning, NLP with Deep Learning, Linear Algebra, Numerical Analysis, Statistical Inference

EXPERIENCE

AI-Integrated Mobile App Developer Intern, EdgeFirm Jul 15 - Aug 31, 2026

- Took over the Weather Ways project mid-cycle and delivered its remaining three sprints across a live three-repository stack (Flutter/GetX, Node.js/Prisma, Next.js) through client User Acceptance Testing.
- Implemented OAuth 2.0 identity-token validation and a Twilio SMS fan-out that dispatches high-severity emergency alerts (e.g. sudden road closures), with global administrative kill-switches.
- Added AWS CloudWatch (backend) and Firebase Crashlytics (mobile) observability to catch silent failures and edge cases in production.
- Rewrote coordinate-parsing logic to fix map rendering artifacts reported by the client, so backend spatial data renders as smooth traffic-flow polylines on mobile.
- Integrated the Gemini API for weather-aware route optimization, managing app state and latency so outputs became navigable routes.

Math & Problem-Solving Instructor, IBAGrads (IBA and SAT entry-test prep institute), Karachi Jul 2024 - Apr 2026

- Taught about 70 students across four batches, plus 4 private students; selected after ranking 1st in the IBA entry test, student now enrolled across different programs at IBA and abroad universities.

PROJECTS

PantryPal: Full-Stack Meal Planning Platform (course project, 3 members; Lead Architect & Developer) | *React 19, Node.js, Express, PostgreSQL (Supabase)*

- Architected a full-stack inventory and recipe-discovery app with 40+ REST endpoints, JWT authentication, multi-filter search and multipart uploads via Multer.
- Designed the relational schema and wrote PL/pgSQL stored procedures for real-time pantry deductions and daily macro aggregation.
- Built a drag-and-drop weekly meal calendar and a real-time cooking countdown tracker.

AcademIQ: Academic Resource Platform (term project, 4 members; Project Lead & Full-Stack Developer) | *JavaScript, Node.js, SQL*

- Led a four-person team through Agile sprints, managing feature branching and splitting work between UI and database development.
- Built email/phone authentication, an interactive calendar module and structured resource hubs, scoped from a survey of IBA students.

Multi-Agent Path Finding Engine (solo) | *Python, PyTorch*

- Built a decentralized path-finding engine for a 25×35 grid with 1-tile aisles, combining A* search with a Constraint Satisfaction framework and using Priority Inheritance with Backtracking to resolve deadlocks.
- Trained a DQN (curriculum learning, 10,000-sample replay buffer) reaching 100% success across six deterministic edge cases.

SKILLS

- **Languages:** JavaScript, Python, C++, SQL, HTML/CSS
- **Frontend:** React, React Router, Vite, Tailwind
- **Backend & Data:** Node.js, Express, Prisma, PostgreSQL (Supabase), Firebase, REST APIs, JWT, OAuth 2.0
- **Cloud & Tools:** AWS CloudWatch, Twilio, Git/GitHub, VS Code, Cursor, Claude Code, MS Entra, MS Power Apps, Trello

LEADERSHIP

Module Head, Enigma, IBA's flagship event 2023, 2024

- Led a 20-person team running a module for 16 competing teams (144 participants); selected two years running.

CERTIFICATIONS

- Supervised Learning with scikit-learn – DataCamp
- Cluster Analysis in Python – DataCamp
- Introduction to MLflow – DataCamp
- AI for Professionals – LUMSX/ILMx
- Introduction to Git – DataCamp (In Progress)
- Business Communication – ILMx